

Manufacturing Productivity

Analyzing factory efficiency.



Our Team



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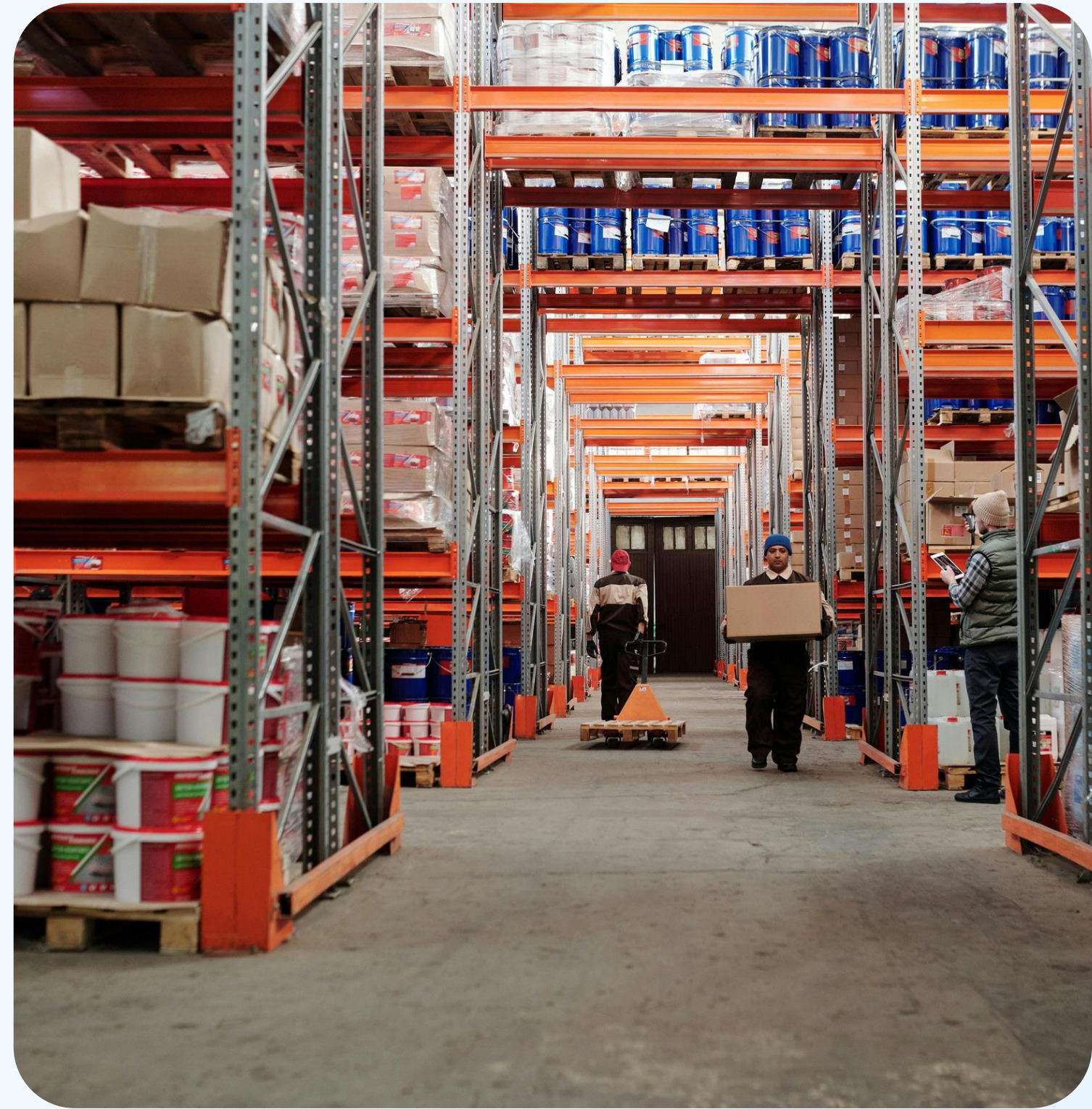
Executive Summary

Manufacturing Downtime Analysis
Insights – Forecasting – Productivity



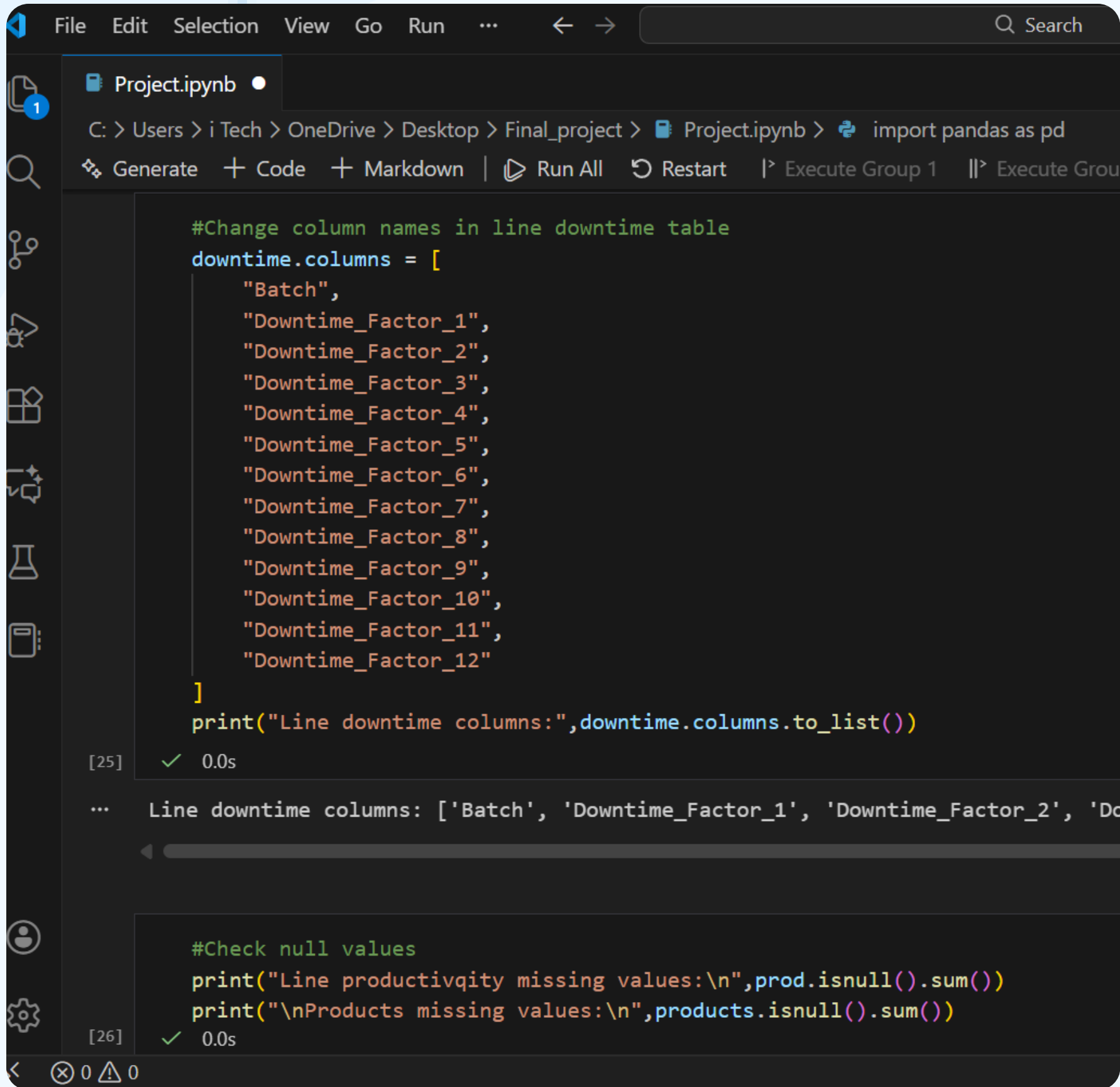
Introduction

Manufacturing downtime disrupts production and lowers efficiency. By analyzing downtime events and daily output, we can identify the main causes and improve performance.



Methodology

- 1 We cleaned the data using Python by fixing missing values, correcting errors, and preparing it for analysis.



```
File Edit Selection View Go Run ... Search
Project.ipynb
C: > Users > i Tech > OneDrive > Desktop > Final_project > Project.ipynb > import pandas as pd
Generate + Code + Markdown | Run All Restart |> Execute Group 1 ||> Execute Group 2

#Change column names in line downtime table
downtime.columns = [
    "Batch",
    "Downtime_Factor_1",
    "Downtime_Factor_2",
    "Downtime_Factor_3",
    "Downtime_Factor_4",
    "Downtime_Factor_5",
    "Downtime_Factor_6",
    "Downtime_Factor_7",
    "Downtime_Factor_8",
    "Downtime_Factor_9",
    "Downtime_Factor_10",
    "Downtime_Factor_11",
    "Downtime_Factor_12"
]
print("Line downtime columns:",downtime.columns.to_list())

[25] ✓ 0.0s
... Line downtime columns: ['Batch', 'Downtime_Factor_1', 'Downtime_Factor_2', 'Downtime_Factor_3', 'Downtime_Factor_4', 'Downtime_Factor_5', 'Downtime_Factor_6', 'Downtime_Factor_7', 'Downtime_Factor_8', 'Downtime_Factor_9', 'Downtime_Factor_10', 'Downtime_Factor_11', 'Downtime_Factor_12']

#Check null values
print("Line productivity missing values:\n",prod.isnull().sum())
print("\nProducts missing values:\n",products.isnull().sum())

[26] ✓ 0.0s
< 0 0 0
```


AutoSave Off project googlesheet updated.xlsx - Protected... • Saved to this PC

Home WPS PDF Insert Draw Page Layout Formulas Data Review View Automate Help Power Pivot

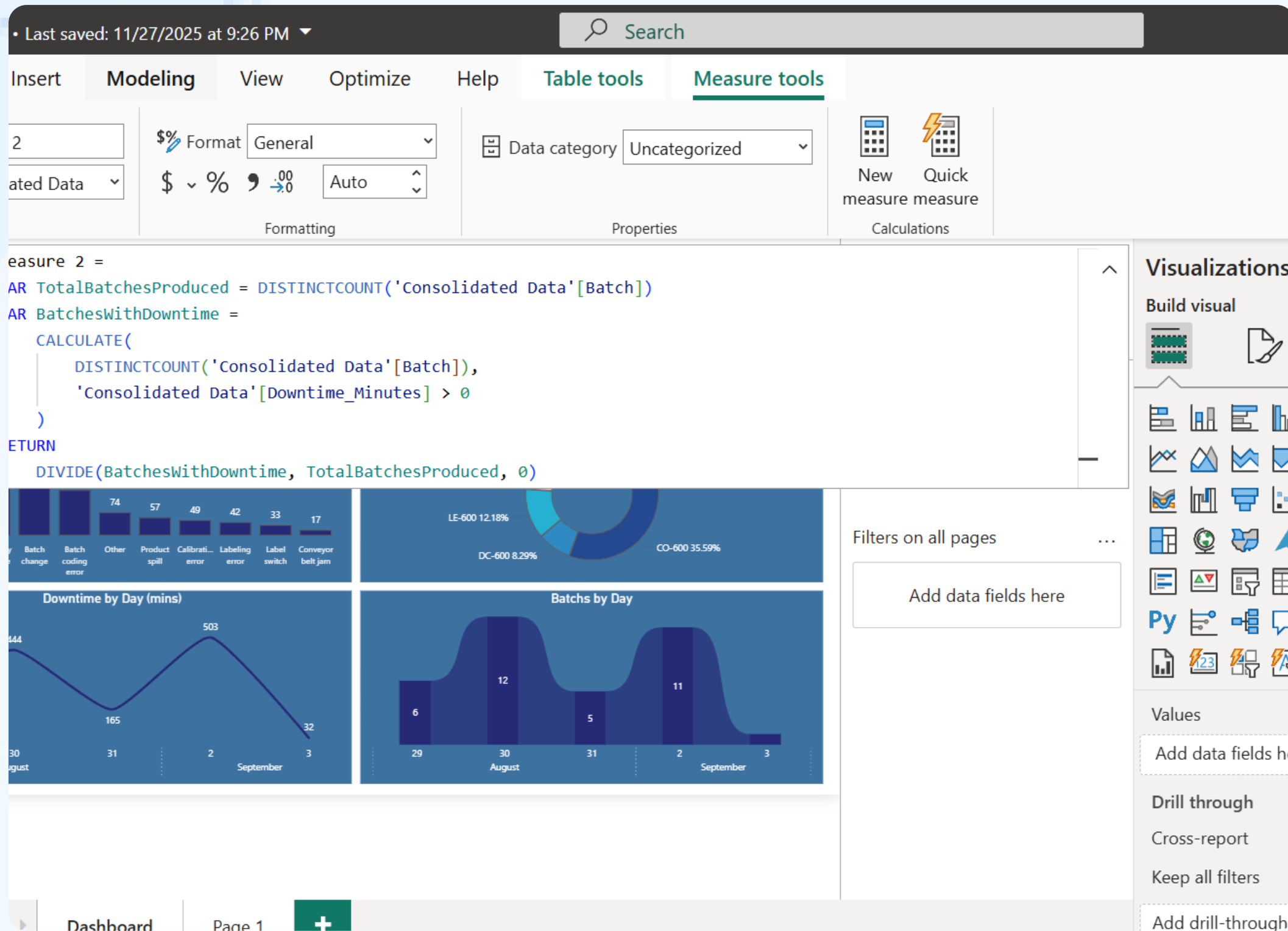
PROTECTED VIEW Be careful—files from the Internet can contain viruses. Unless you need to edit, it's safer to stay in Protected View. Enable Editing

fx =SUM(N2:N40)

E	F	G	H	I	J	K	L	M	N
0	0	5	0	0	0	0	0	0	15
0	0	14	16	0	0	0	10	20	60
25	0	0	0	0	0	0	0	0	25
20	15	0	0	0	17	0	0	0	52
0	0	0	15	0	0	0	0	0	15
0	0	0	25	0	0	0	0	0	25
43	0	0	30	0	0	0	0	0	73
0	20	20	0	0	0	0	0	0	40
0	0	0	0	0	0	0	10	10	20
0	0	0	0	44	0	0	0	0	44
0	0	23	0	0	0	0	0	0	23
0	22	0	30	0	0	0	0	0	52
0	0	0	0	0	0	0	0	15	15
0	0	0	0	0	0	0	0	0	20
20	0	0	0	0	0	10	0	0	30
0	0	0	0	0	0	0	0	0	0
0	0	0	20	0	0	0	0	0	20
0	0	0	30	20	0	0	0	0	50
30	0	0	0	0	0	0	0	15	45
0	0	0	0	0	0	0	0	0	0
0	0	0	0	30	0	15	0	0	45
0	0	0	0	0	0	0	0	0	20
20	0	15	0	0	0	0	0	0	35
0	0	50	0	0	0	0	13	0	63
0	0	0	0	0	0	0	0	7	7
0	0	30	0	0	0	0	0	0	30
0	0	40	18	0	0	0	0	0	58
0	0	30	0	24	0	0	0	0	54
0	0	0	0	0	0	0	0	0	22
0	0	30	25	0	0	0	0	7	62
17	0	60	30	0	0	0	0	0	107
25	0	0	0	7	0	0	0	0	32
225	57	332	254	145	17	49	33	74	1388

Cleaned_Products Cleaned_Downtime_Factors Cleaned_Line_Productivity Cleaned_Line_Downtime Copy of KPIs +

- 2 We calculated the main KPIs and performance metrics using Excel to understand production behavior and downtime impact.



3 We created an interactive dashboard using Power BI to visualize downtime patterns, operator performance, and daily batch output.



Results



Manufacturing Productivity

Operator

Charlie

Dee

Dennis

Mac

Product

All

Date

8/29/2024

9/3/2024

56%

% Operator Error

23 h 8 m

Total Downtime

1h 52m

Avg Production Duration

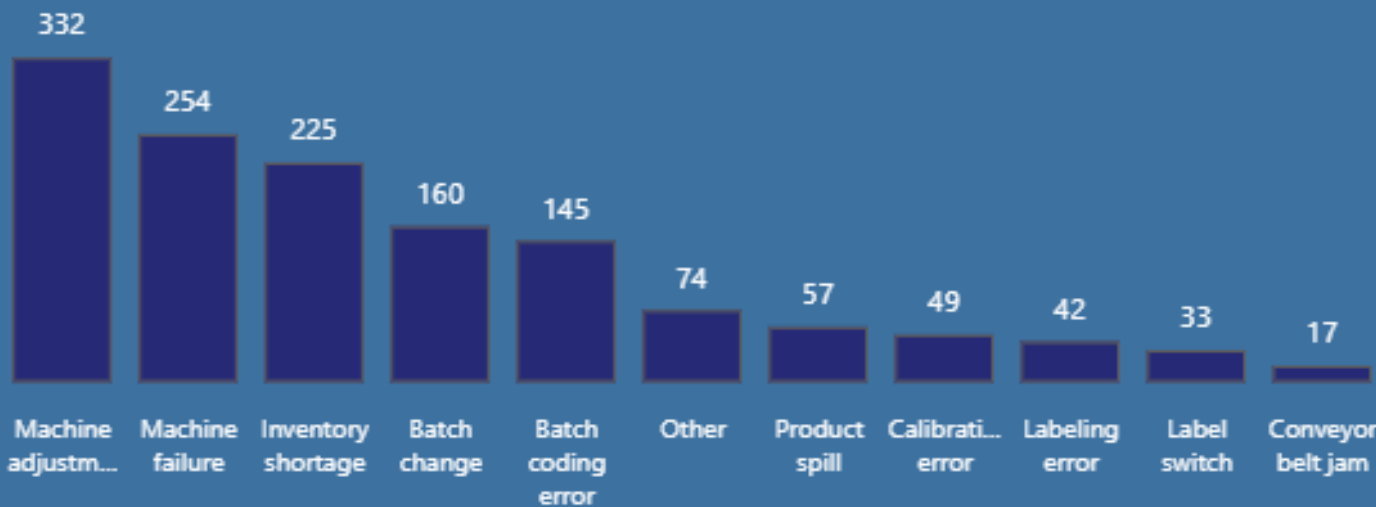
23 m

Avg Occurrence

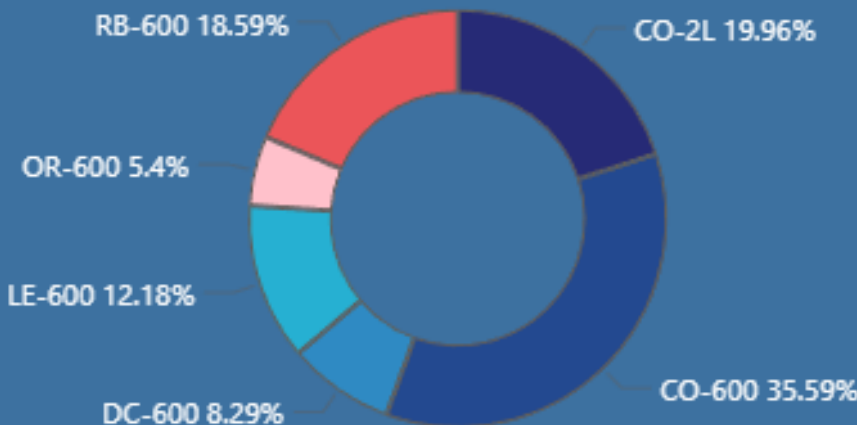
40 m

Avg Downtime per Batch

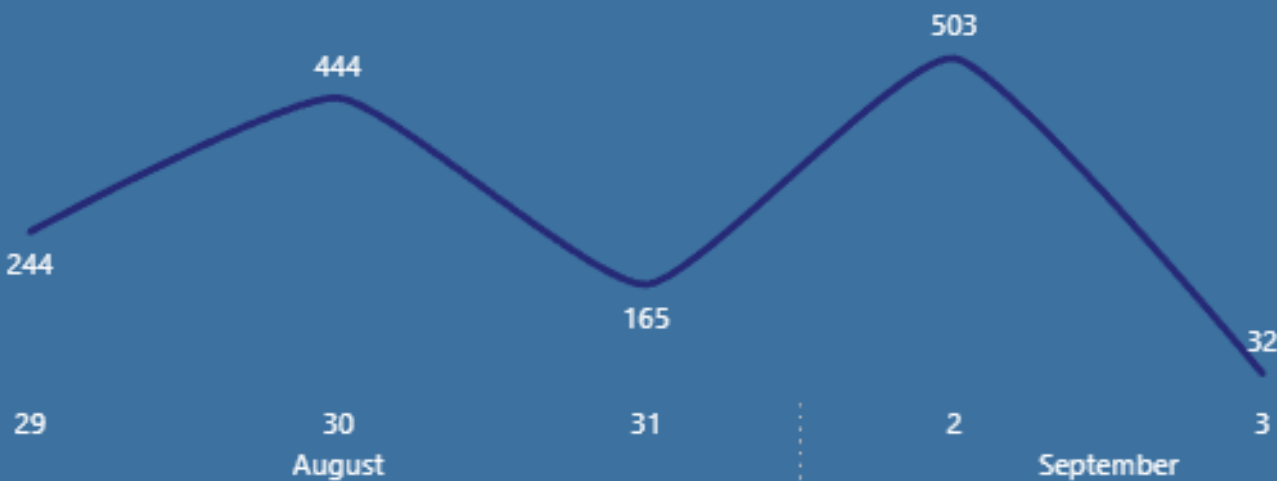
Downtime by Factor (mins)



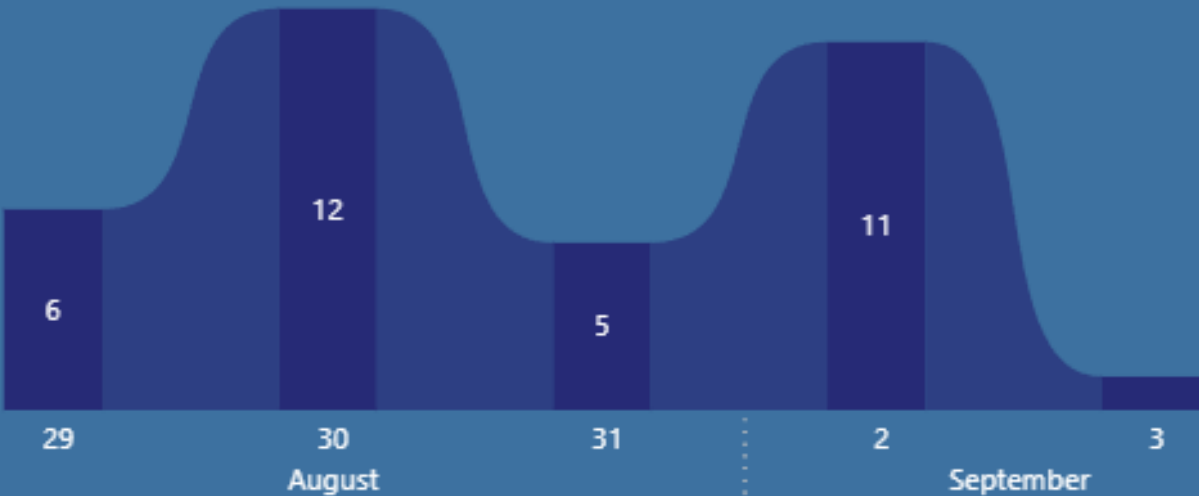
Downtime by Product (%)



Downtime by Day (mins)



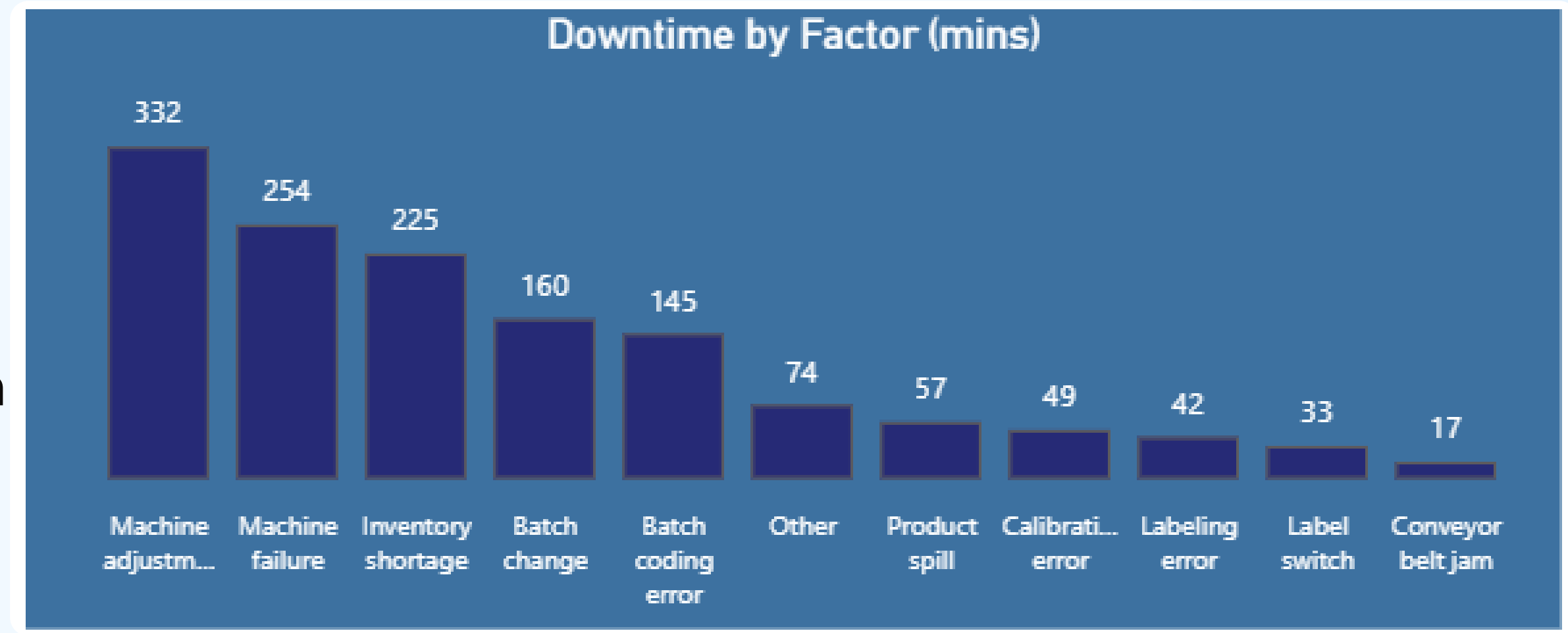
Batches by Day





✓ Total Downtime by Factor

The chart shows that machine adjustment and machine failure are the main causes of downtime, while other factors have a smaller impact.





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23 h 8 m

Total Downtime

1h 52m

Avg Production Duration

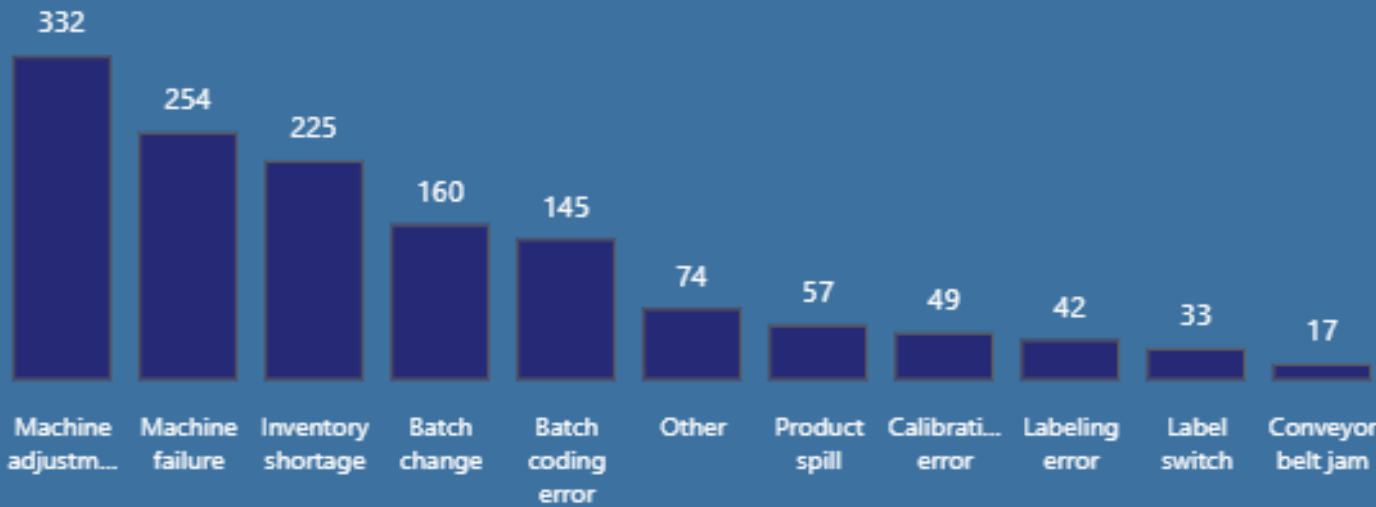
23 m

Avg Occurrence

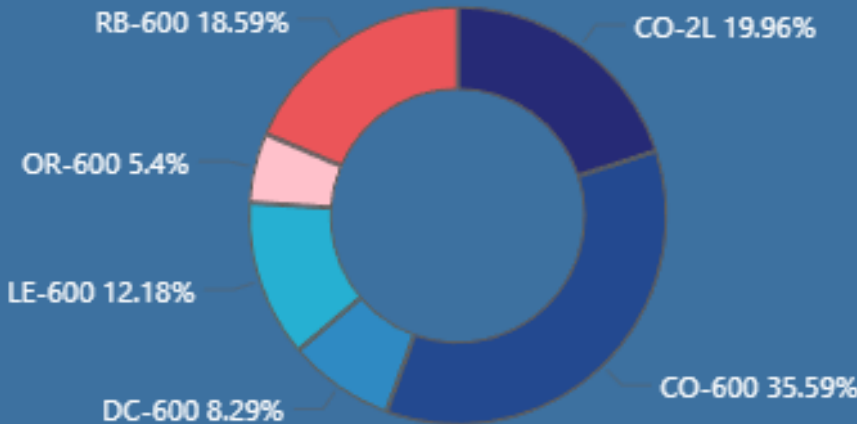
40 m

Avg Downtime per Batch

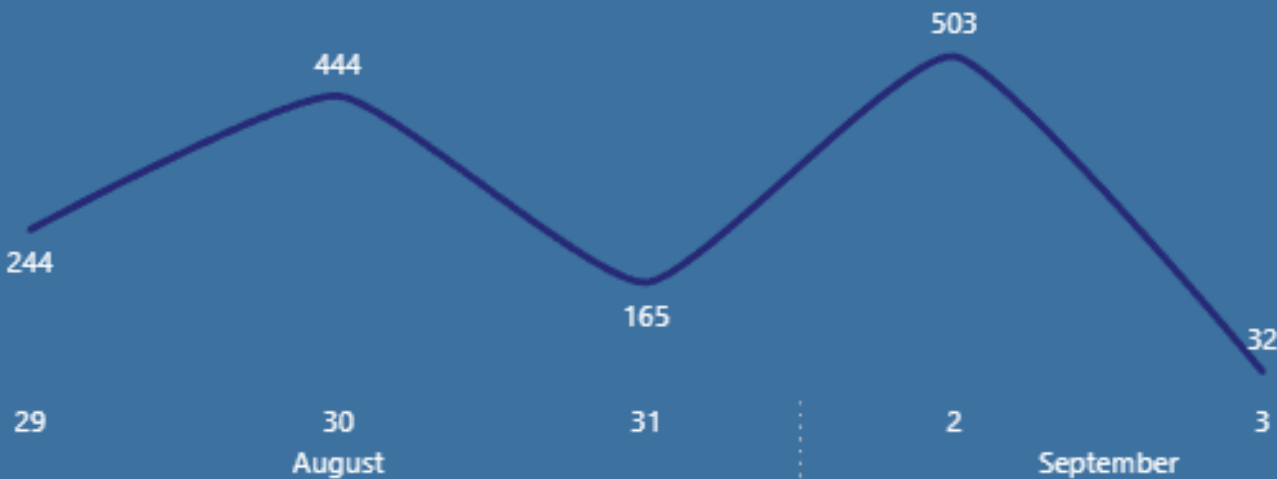
Downtime by Factor (mins)



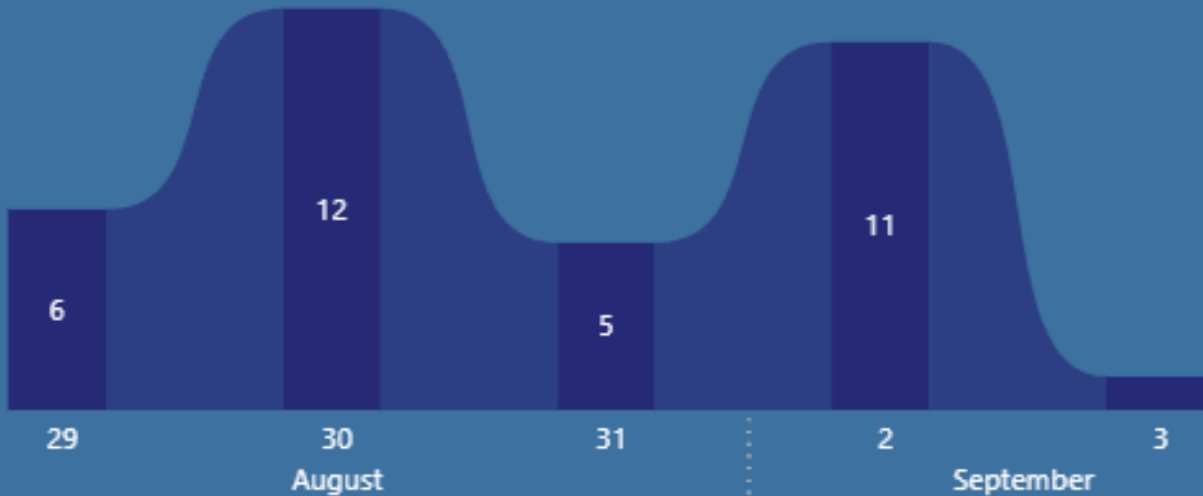
Downtime by Product (%)



Downtime by Day (mins)



Batches by Day

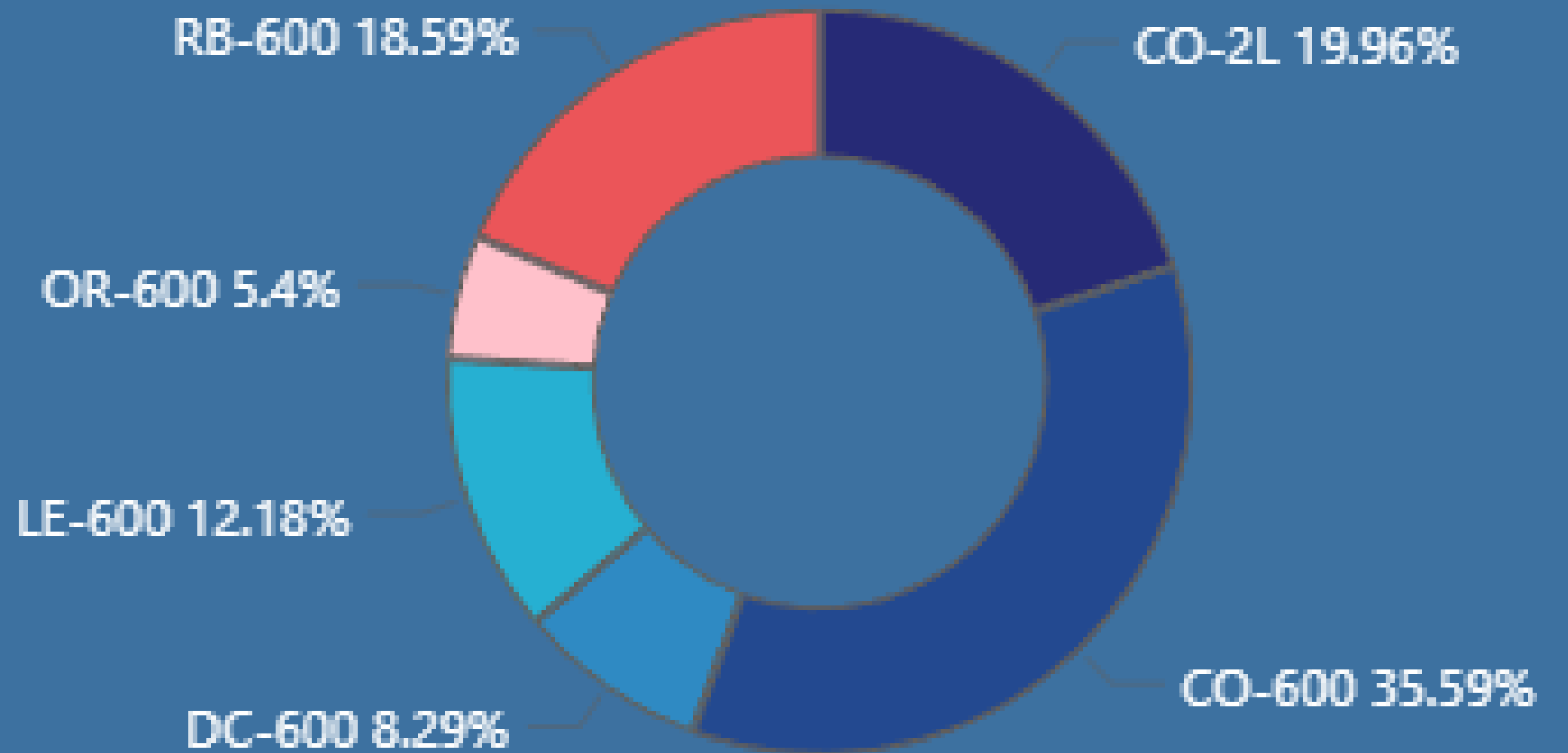




✓ Downtime by Product

The chart shows that CO-600 has the highest downtime, while products like OR-600 and DC-600 have the lowest.

Downtime by Product (%)





Results



Manufacturing Productivity

Operator

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Dee

Dennis

Mac

Product

All

Date

8/29/2024

9/3/2024

56%

% Operator Error

23 h 8 m

Total Downtime

1h 52m

Avg Production Duration

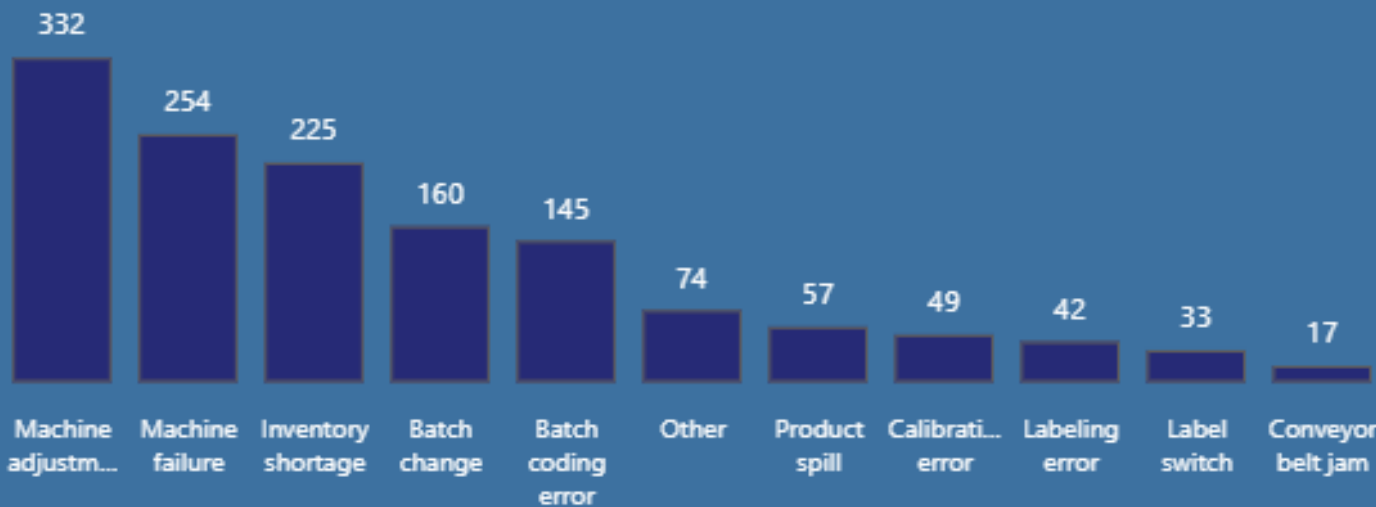
23 m

Avg Occurrence

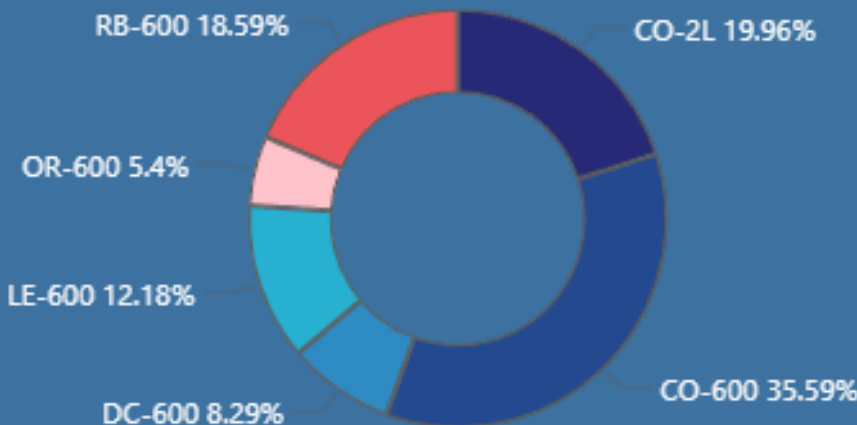
40 m

Avg Downtime per Batch

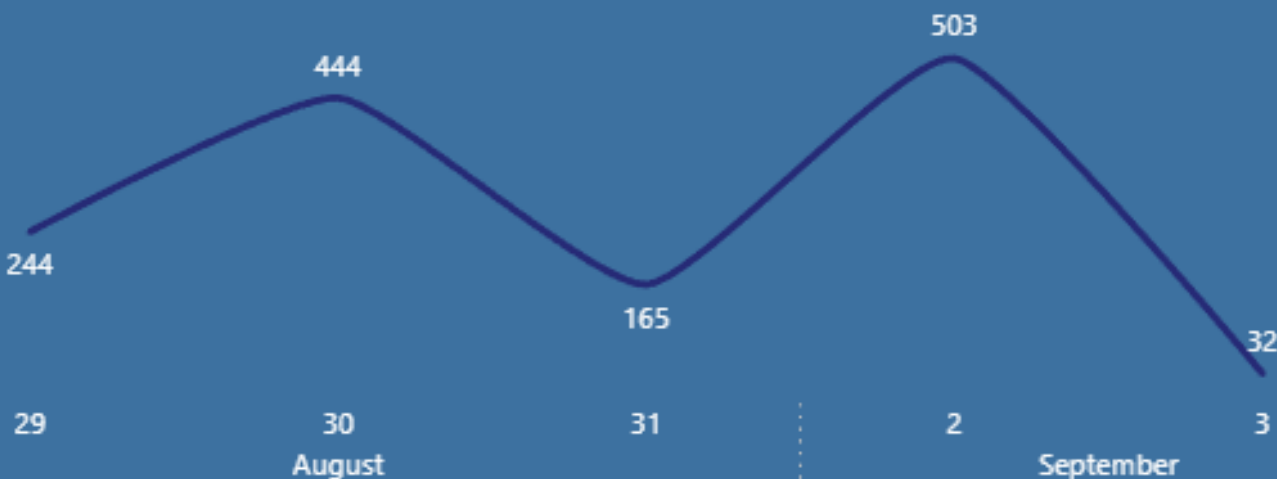
Downtime by Factor (mins)



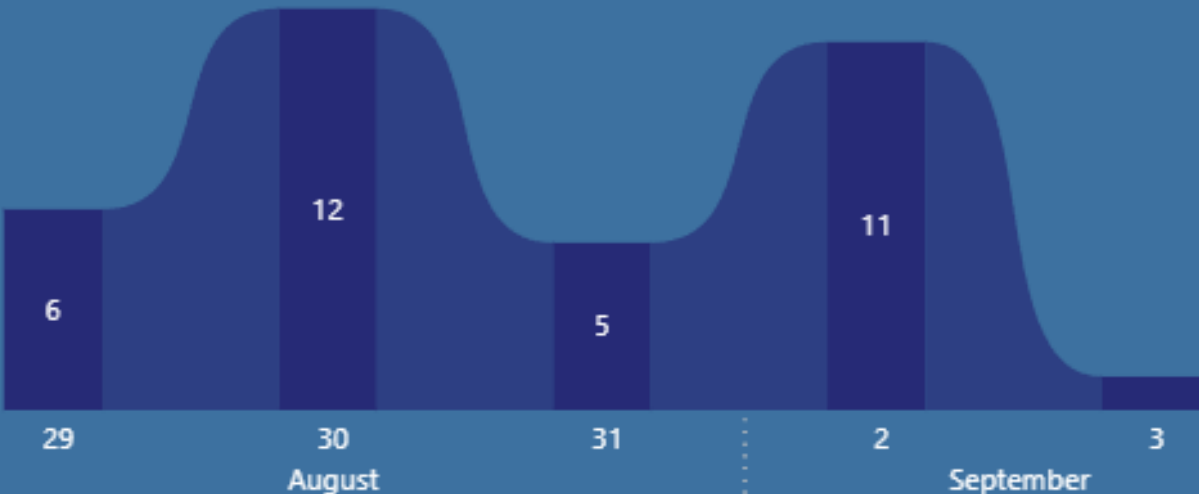
Downtime by Product (%)



Downtime by Day (mins)



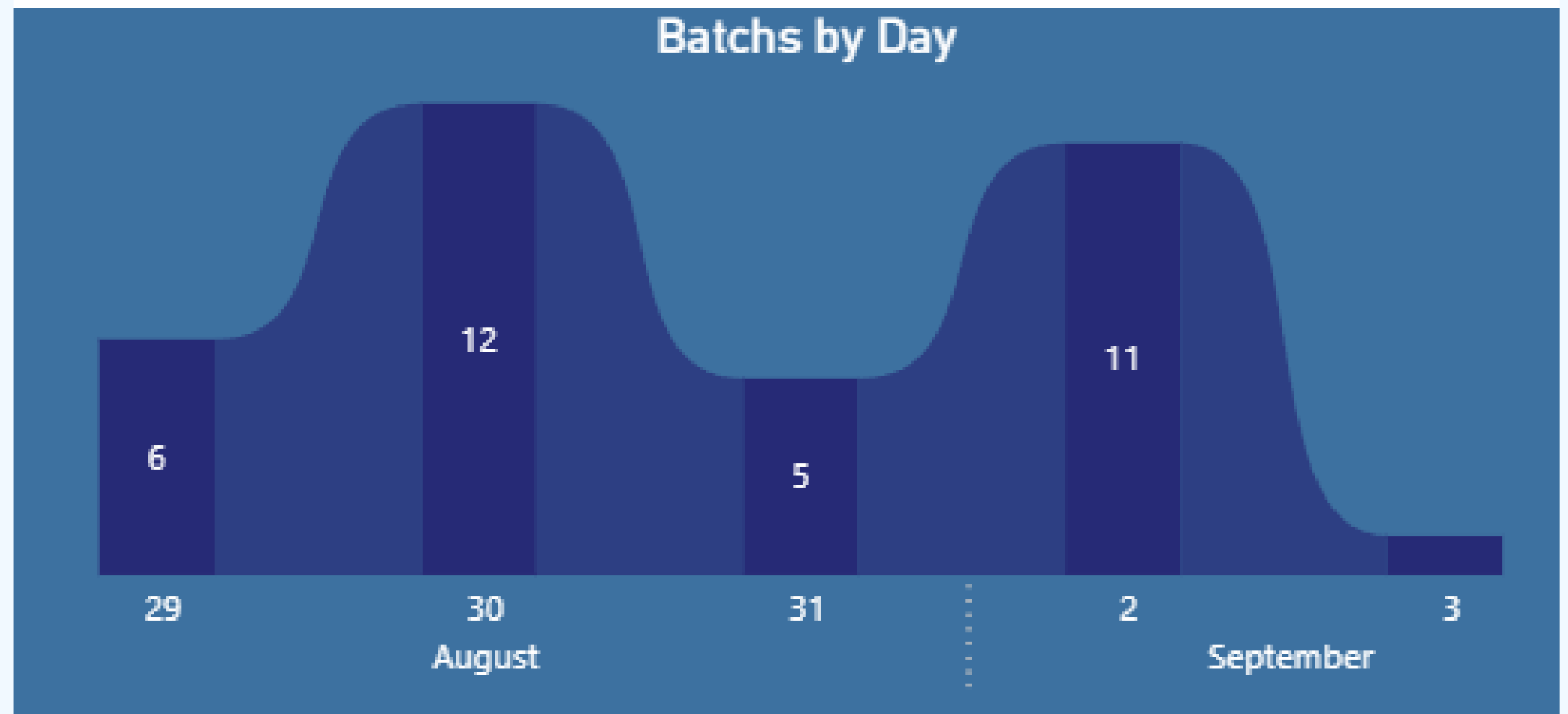
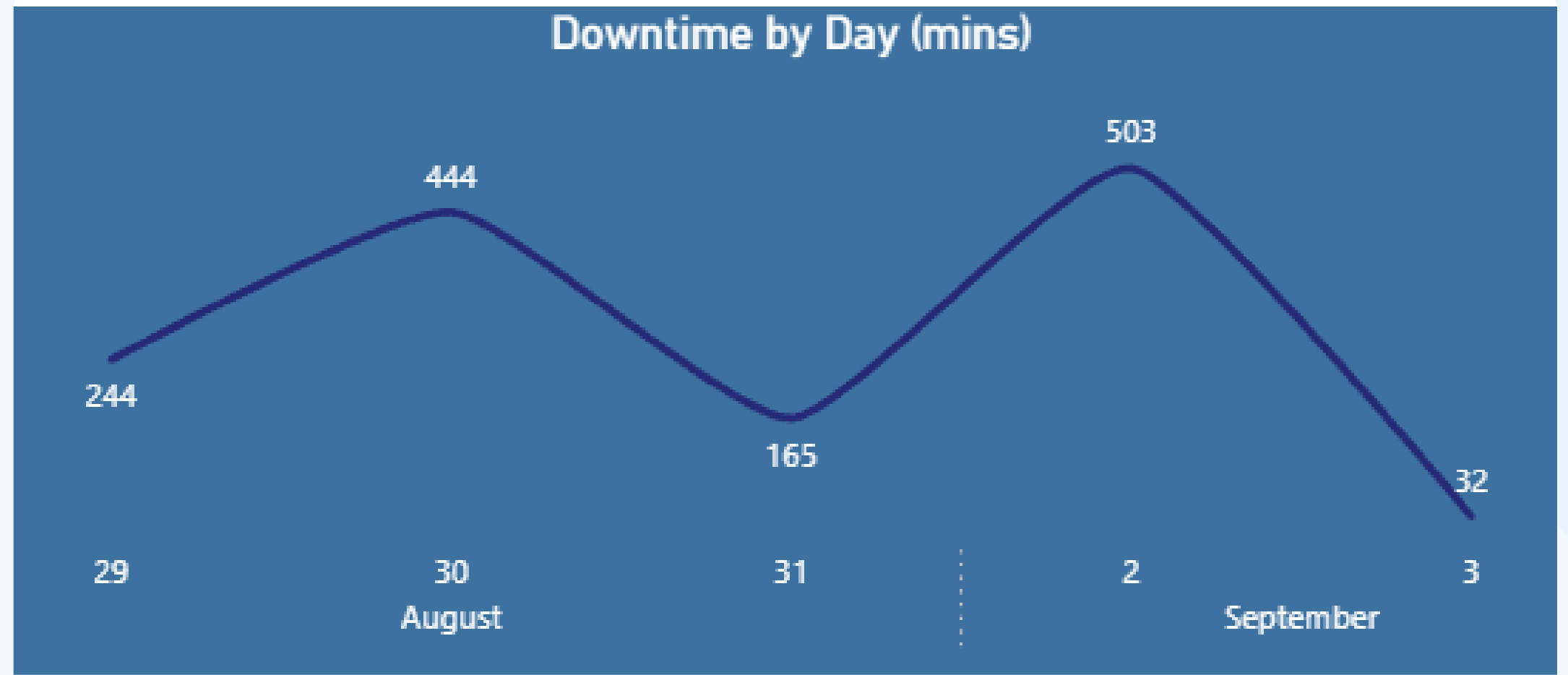
Batches by Day





Downtime & Batches Produced by Day

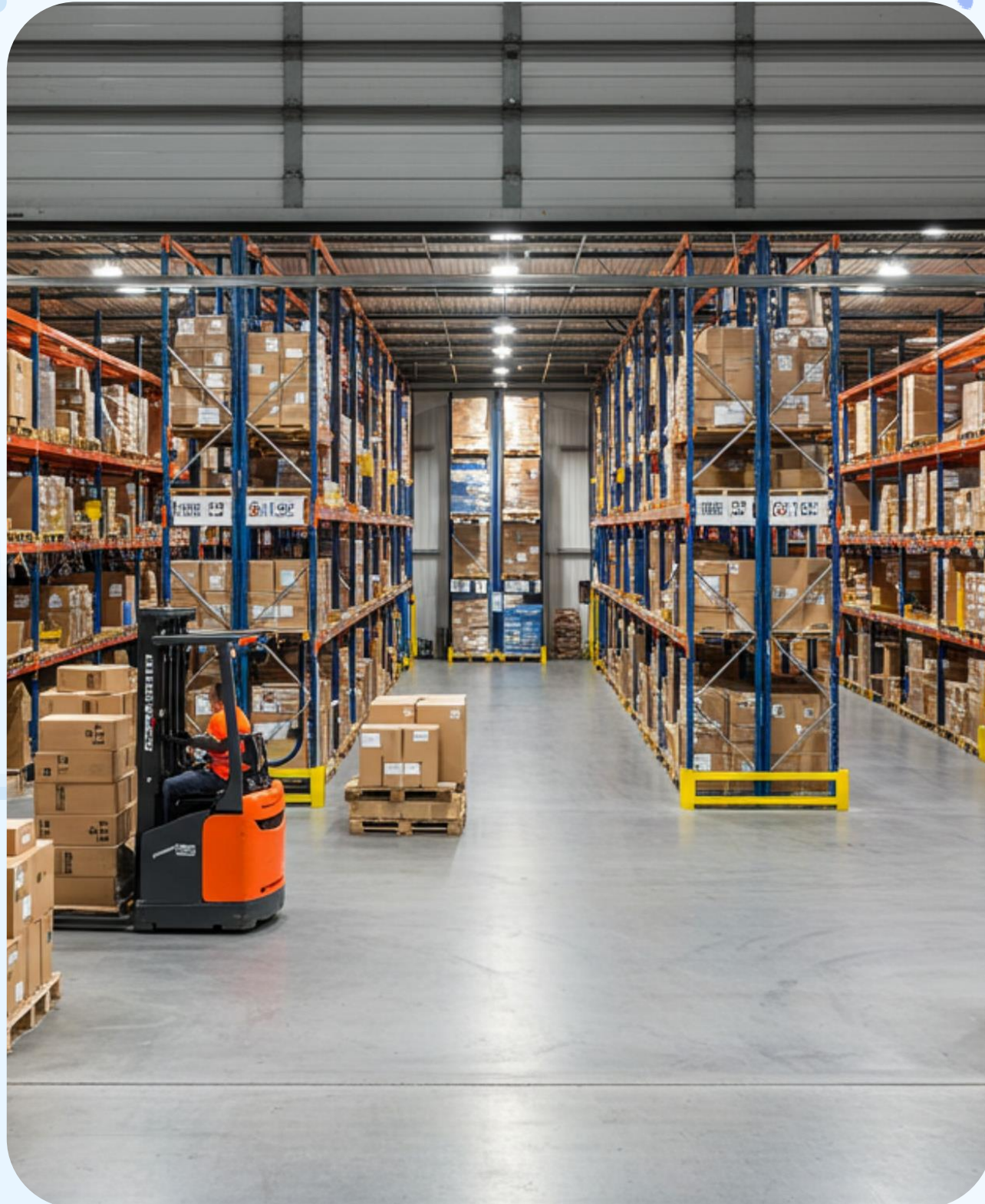
Higher downtime days match higher production activity, while very low downtime (like September 3) shows minimal batches produced.



Conclusion ✕

- ✓ Operator errors are the main source of downtime, indicating process instability.
- ✓ Total downtime (23h) reflects significant productivity loss.
- ✓ Frequent interruptions show a system-wide performance issue.
- ✓ Machine failures remain a key technical bottleneck.
- ✓ CO-600 is the most affected product, creating concentrated inefficiency.
- ✓ Large daily fluctuations reduce consistency and predictability.





Recommendations

- ✓ Eliminate operator-related errors by enforcing strict training, clear SOPs, and continuous performance monitoring.
- ✓ Adopt a preventive maintenance strategy to reduce machine failures and ensure consistent line availability.
- ✓ Stabilize the production workflow through standardized processes, better scheduling, and tighter control over daily operations.
- ✓ Prioritize optimization of the CO-600 line, as it creates the largest productivity loss, requiring focused root-cause analysis and corrective action.



**Thank You
for Listening.**